Table 1: Leave-one-out ablation of the two core contributions on CORE4D-G1 (all 50 paper sequences). Each ablation removes one sub-component and re-runs the sampling-based MPC; **Ours** is the full method. *Contribution 1 (Multi-Scale Contact)*: w/o local reward removes the fine SDF surface-band term, w/o global reward removes the coarse anchor-attraction term. *Contribution 2 (Part-wise Penetration)*: w/o hard select removes all hard candidate gates, w/o soft penalty removes all soft penetration penalties. Physical contact / penetration are reported at both 5 mm and 10 mm bands; MPJPE is the whole-body mean per-joint Cartesian error (MPJPE-G global, MPJPE-L pelvis-aligned). ↑/↓: higher/lower is better; **bold** = best, underline = second best. Note w/o global reward's lower penetration is an artifact of failing to reach the object (see MPJPE-G), not better contact.

Metric	Ours	w/o local reward	w/o global reward	w/o hard select	w/o soft penalty
Phys. contact@5mm (%) ↑	**72.5**	39.6	47.2	70.0	71.6
Phys. contact@10mm (%) ↑	**80.2**	54.0	50.8	79.2	80.2
Phys. penetration@5mm (%) ↓	7.6	12.6	**3.6**	8.7	8.5
Phys. penetration@10mm (%) ↓	1.7	2.8	**0.9**	1.6	2.0
Max phys penetration (mm) ↓	12.0	16.1	**10.6**	13.1	12.8
MPJPE-G (cm) ↓	14.34	**12.28**	40.07	18.88	15.78
MPJPE-L (cm) ↓	4.55	3.64	**2.99**	6.87	5.07
Obj pos err (cm) ↓	10.34	**9.74**	10.24	10.74	10.48
Obj ori err (deg) ↓	5.11	4.99	**4.97**	5.47	5.60